C. Sequence of points

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given the following points with integer coordinates on the plane: $M_0, A_0, A_1, \ldots, A_{n-1}$, where n is odd number. Now we define the following infinite sequence of points M_i: M_i is symmetric to M_{i-1} according $A_{(i-1) \bmod n}$ (for every natural number i). Here point B is symmetric to A according M, if M is the center of the line segment AB. Given index j find the point M_j.

Input

On the first line you will be given an integer n ($1 \le n \le 10^5$), which will be odd, and j ($1 \le j \le 10^{18}$), where j is the index of the desired point. The next line contains two space separated integers, the coordinates of M_0. After that n lines follow, where the i-th line contain the space separated integer coordinates of the point A_{i-1}. The absolute values of all input coordinates will not be greater then 1000.

Output

On a single line output the coordinates of M_j, space separated.

Examples

input Copy

```
3 4
0 0
1 1
2 3
-5 3
```

output Copy

```
14 0
```

input Copy

```
3 1
5 5
1000 1000
-1000 1000
3 100
```

output Copy

```
1995 1995
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 24

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372136136	Apr/22/2026 15:59	Accepted
372135363	Apr/22/2026 15:53	Accepted
372135101	Apr/22/2026 15:50	Accepted

→ Problem tags

geometry implementation math *1800

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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The only programming contests Web 2.0 platform
Server time: Apr/22/2026 20:00:28 UTC+7 (n2).
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